

NELSON KOSKELA

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience in Python and production data infrastructure, with hands-on experience building and shipping agentic AI applications. Designed a live SDLC visualizer and built a RAG AI assistant for my personal website. Experienced in translating business requirements into technical architecture and production solutions through Agile practices and direct customer communication on time-sensitive projects. Independently designed and built a data warehouse with an automated test suite and adversarial review process, applying dimensional modeling principles to data and AI systems.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java, C#, Swift, HTML/CSS

AI & Agentic Tooling: Agentic engineering, AI-agent orchestration, AI-assisted development workflows

Data & Analytics: dbt, dimensional/Kimball modeling, DuckDB, MotherDuck, PostgreSQL, SQLite3, Athena (J.P. Morgan Chase internal data system)

Frameworks & Libraries: Flask, React, Bootstrap, Ruby on Rails

Cloud & Infrastructure: AWS, Docker, Redis, Unix/Linux, Git

Methodologies: Agile/Scrum, Object-Oriented Programming, Distributed Systems, Data Reconciliation

PROFESSIONAL EXPERIENCE

J.P. Morgan Chase & Co. - Software Engineer II, Corporate & Investment Banking *July 2022 - September 2026*

Houston, TX

- Engineered and operated data pipelines and analytics infrastructure supporting regulatory and risk reporting, reliably processing multi-terabyte datasets across global business units in a production, always-on environment.
- Partnered directly with business stakeholders to define requirements and build new reporting and infrastructure for enterprise risk-data systems, delivering data in live, end-of-day (EOD), and start-of-day (SOD) formats to meet real decision-making needs.
- Stepped up as de facto project lead on a full system migration amid a team staffing shortage, owning prioritization of downstream requests, cross-team coordination, and bug triage on a tight timeline.
- Designed and led a parallel "prod vs. prod" validation framework end to end, with automated reconciliation ensuring the two environments matched to the record before cutover.
- Applied agentic engineering to analyze large volumes of data and conduct deep dives into the Athena ecosystem, cutting support workload by 50%.
- Partnered with downstream and cross-functional Agile teams to enforce data quality, security, and regulatory compliance, operating within strict access-control, auditability, and governance requirements.

StreetCode Academy - Diversity & Inclusion Officer & Freelance Developer

Fall 2020 - Fall 2022

Remote

- Developed responsive web applications using HTML, CSS, and JavaScript, enhancing customer engagement and usability.
- Applied Agile Scrum methodology to plan sprints, track progress, and deliver client solutions on time.
- Managed logistics and database tracking systems for internal inventory, improving item tracking accuracy by 50% by building and delivering full, accurate analytics reporting.

PROJECTS

Market Data Warehouse - nelsonkoskela.dev/projects/market-warehouse

- Designed and built a Kimball-style dimensional data warehouse (dbt, 3 conformed dimensions, 5-table fact family, SCD-1/SCD-2) from a live external market-data API, portable across DuckDB, MotherDuck, and Snowflake with no engine-specific code paths.
- Modeled quant and risk metrics as tested dbt models, including Sharpe ratio, Sortino ratio, beta, volatility, and a three-method price-projection fact (OLS trend, random-walk-drift, AR(1) mean-reversion), backed by 85 automated data-quality tests.

RAG AI Assistant - nelsonkoskela.dev

- Independently scoped and built a RAG-based AI assistant for my personal website using LangGraph, PostgreSQL, pgvector, local embeddings, and Groq for LLM generation.
- Built and operate an automated multi-source job discovery pipeline (Adzuna, Greenhouse, Lever, Ashby, and others) that autonomously searches and LLM-scores new postings against my resume and experience on a recurring production schedule, with qualifying matches promoted into an application tracker through a manual review step and a full audit trail.

EDUCATION

Cogswell Polytechnical Institute - B.S. in Computer Science

2019 - 2022

Relevant Coursework: Data Structures, Database Systems, Linux Programming, Software Engineering Methods

Howard University - B.S. in Computer Science (Transferred)

2017 - 2019

Relevant Coursework: Large Scale Programming, Game Design, Affective Computing